ELECTRONICS

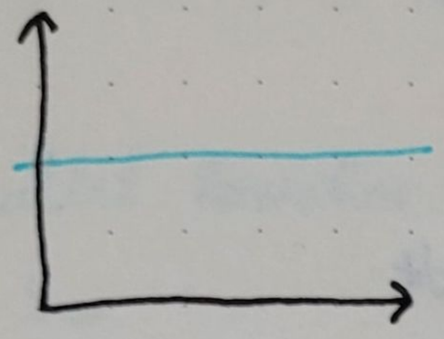
Charge and Current

$$i \triangleq \frac{dq}{dt} \longleftrightarrow Q \triangleq \int_{t_0}^{t} i\, dt$$

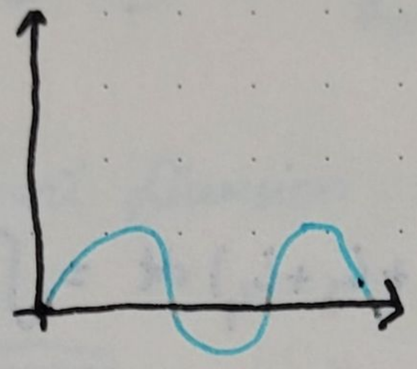
1 Coulomb of electrons $\approx 6 \times 10^{18}$ electrons

1 Coulomb / 1s = 1 A $\quad i = \frac{\Delta Q}{\Delta t}$

DC



AC



Voltage

$$v_{AB} \triangleq \frac{dw}{dq}$$

1 volt = 1 j / coulomb = 1 newton-meter / coulomb

Power

$$p \triangleq \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt}$$

$\frac{dw}{dq} = v$ $\quad \frac{dq}{dt} = I$

$$\int dw = \int p \cdot dt$$

$$w = \int_{t_0}^{t} p \cdot dt = \int_{t_0}^{t} v \cdot i \cdot dt$$

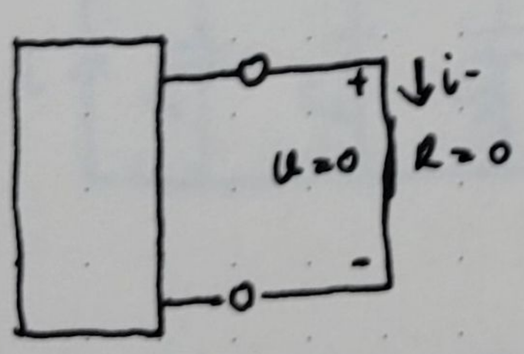
$$\sum p = 0$$

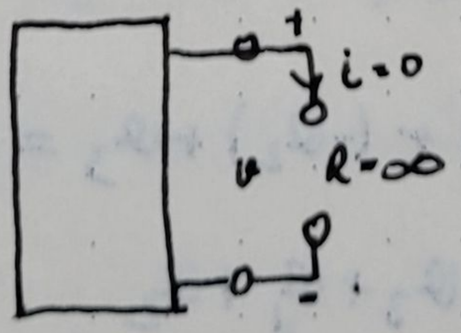
Ohm's Law

$$v = i \cdot R \qquad R = \frac{l \cdot \rho}{A}$$

Carbon, germanium, silicon are semi-conductor.

Short and Open Circuit





$v < 1000$

Branch, Node and Loop

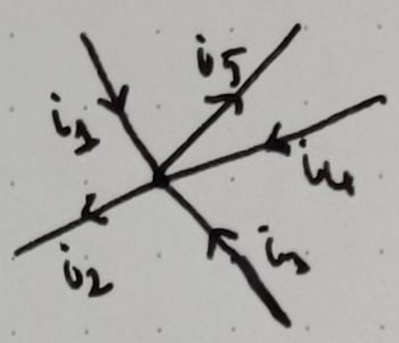
A branch represents a single element such as a voltage source or a resistor.

A node is the point of connection between two or more branches.

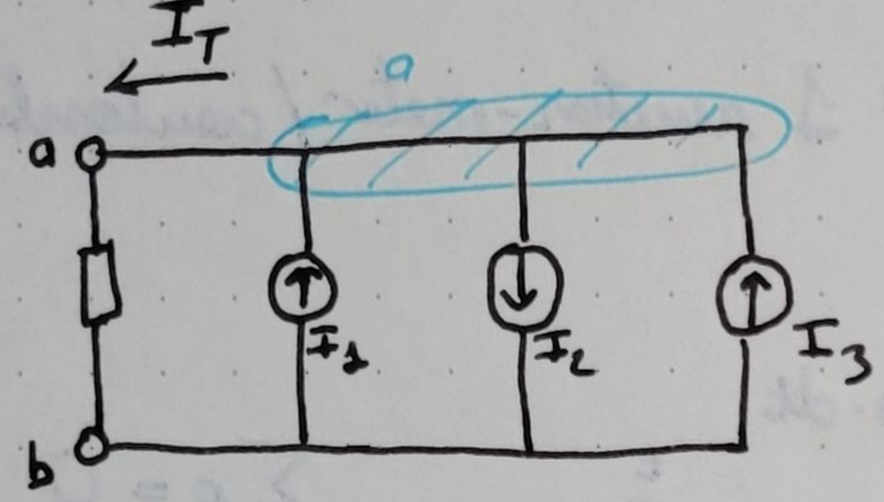
A loop is any closed path in a circuit.

Kirchhoff's Current Law

$$\sum_{n=1}^{N} i_n = 0$$

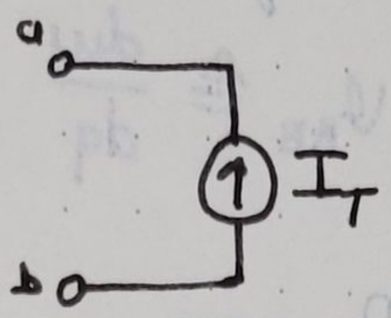


$$\int (i_1 + i_3 + i_4)\,dt = \int (i_2 + i_5)\,dt$$



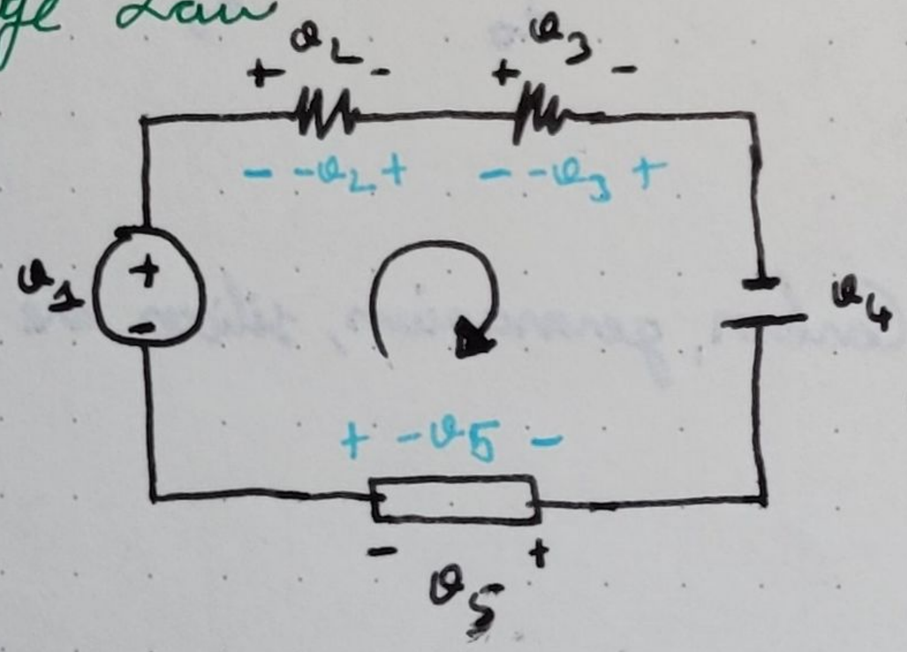
$$I_1 + I_3 = I_2 + I_T$$

$$I_T = I_1 + I_3 - I_2$$



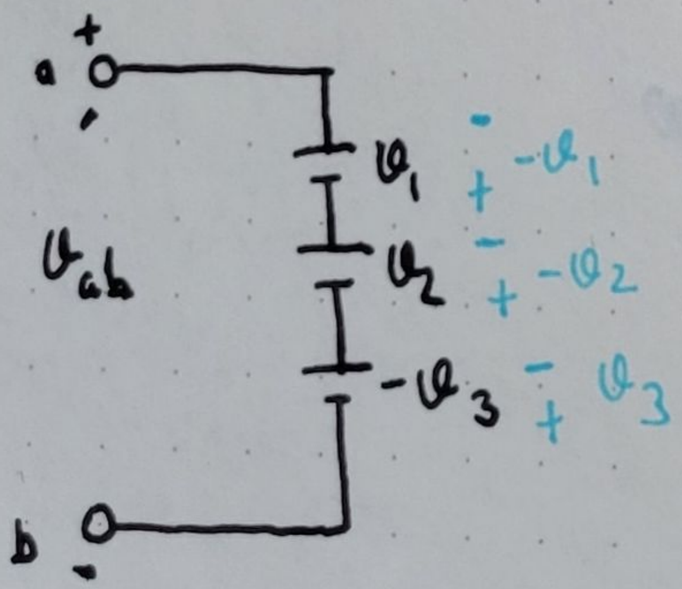
Kirchhoff's Voltage Law

$$\sum_{m=1}^{M} v_m = 0$$



$$v_1 + v_4 = v_2 + v_3 + v_5$$

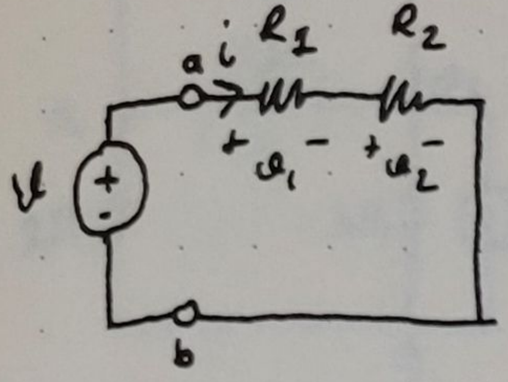
$$v_1 + (-v_2) + (-v_3) + v_4 + (-v_5) = 0$$



$$v_{ab} + (-v_1) + (-v_2) + v_3 = 0$$

$$v_{ab} = -v_3 + v_1 + v_2$$

Series Resistors and Voltage Division

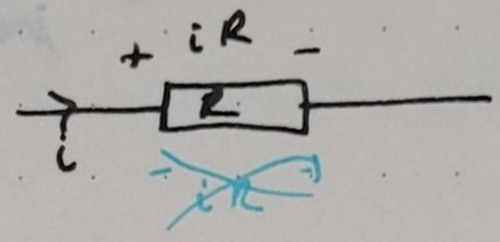


$$R_{eq} = R_1 + R_2$$

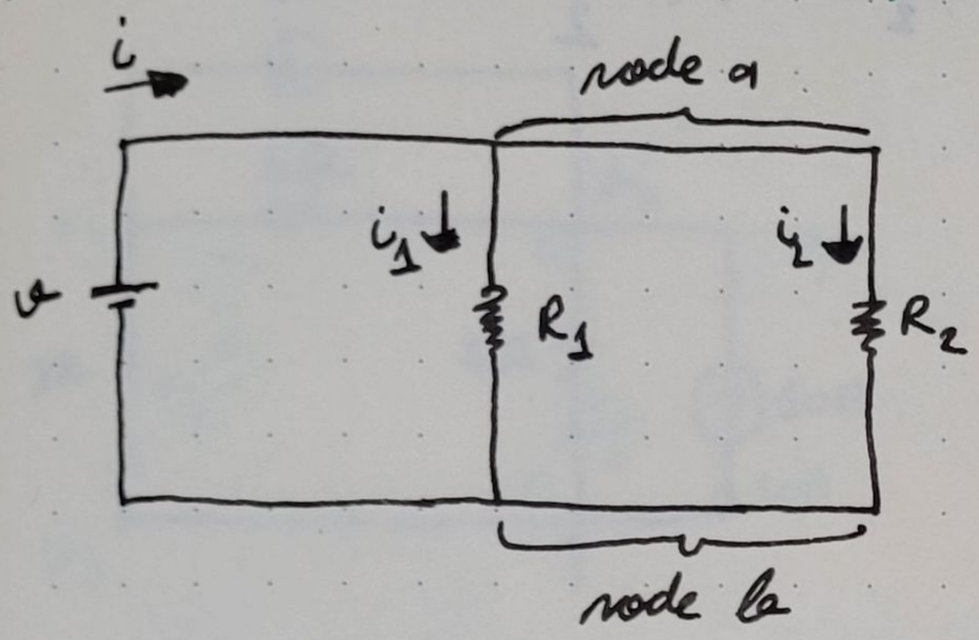
$$v_1 = v \cdot \frac{R_1}{R_2 + R_1}$$

$$v + (-v_1) + (-v_2) = 0$$

$$v = i \cdot R_1 + i \cdot R_2 = i \cdot (R_1 + R_2) \qquad i = \frac{v}{R_1 + R_2}$$



Parallel Resistors and Current Division

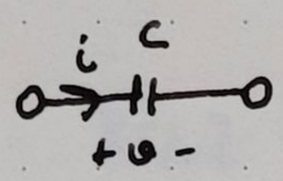


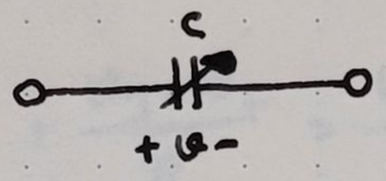
$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \qquad i_1 = \frac{R_2 \cdot i}{R_1 + R_2}, \quad i_2 = \frac{R_1 i}{R_1 + R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$$

Capacitance

$$C = \frac{\varepsilon A}{d} \qquad q = C \cdot V$$

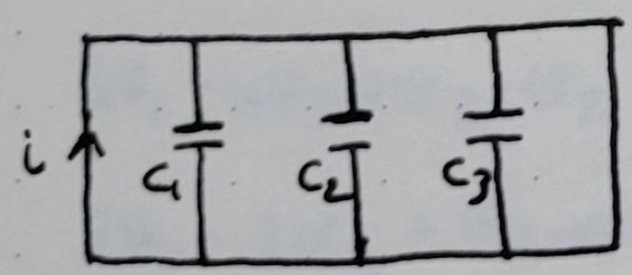




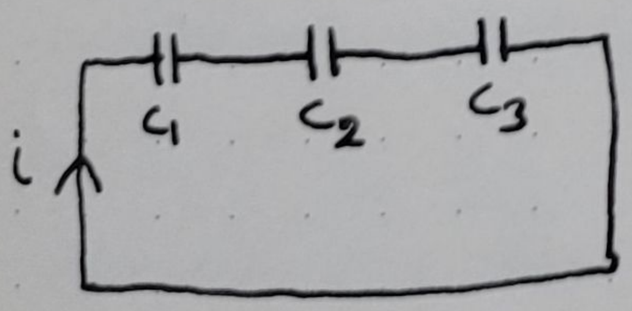
$$i = C \cdot \frac{dv}{dt}$$

Capacitors

$$w = \int_{-\infty}^{t} p(\tau)\, d\tau = C \int_{-\infty}^{t} v \cdot \frac{dv}{d\tau}\, d\tau = C \cdot \int_{v(-\infty)}^{v(t)} v\, dv = \frac{1}{2} C v^2 \Big|_{v(-\infty)}^{v(t)}$$



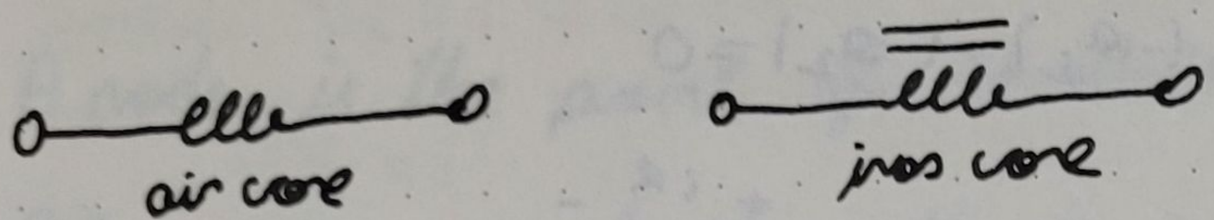
$$C_{eq} = C_1 + C_2 + C_3 + \dots + C_N$$



$$C_{eq} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$

Inductance

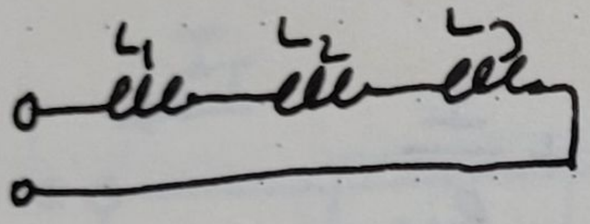
$$L = \frac{N^2 \mu A}{\ell} \qquad v = L \cdot \frac{di}{dt} \rightarrow i = C \cdot \frac{dv}{dt}$$



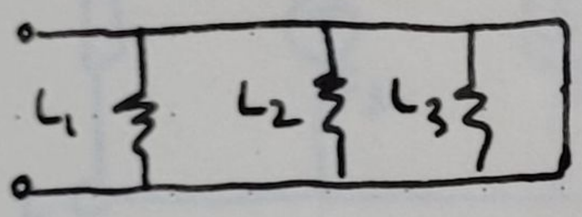
Inductor

$$v = L \cdot \frac{di}{dt} \qquad w = \int_{-\infty}^{t} p(\tau)\,d\tau = L \cdot \int_{-\infty}^{t} \frac{di}{d\tau}\, i\,d\tau = L \int_{-\infty}^{t} i\,di = \frac{1}{2} L \cdot i^2(t) - \frac{1}{2} L i^2(-\infty)$$

$$w = \frac{1}{2} L i^2$$



$$L_{eq} = L_1 + L_2 + \cdots L_N$$

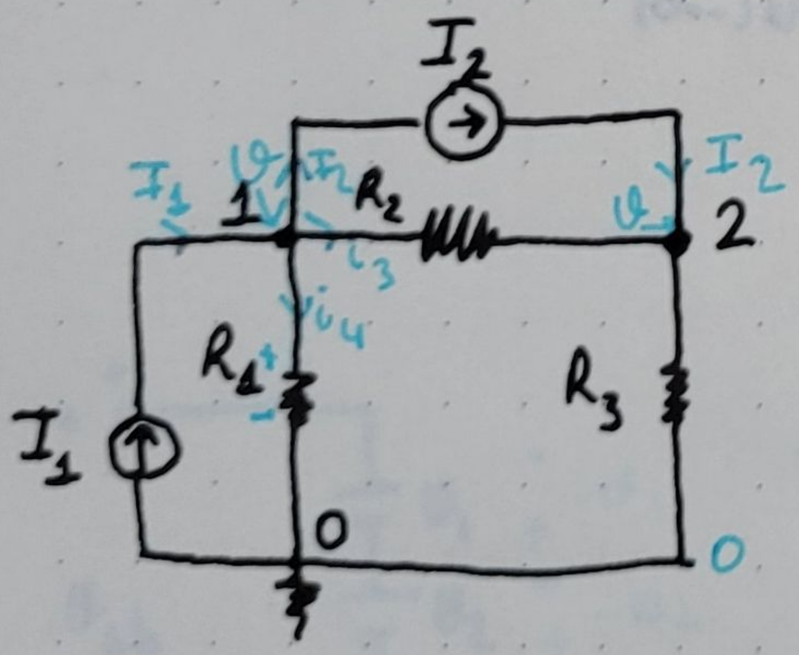


$$L_{eq} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_N}$$

21/03/2024

Nodal Analysis

1. Select a node as the reference node. Assign $v_1, v_2, \cdots, v_{n-1}$
2. Apply KCL to each n-1 non reference node. Use Ohm's law to express branch curr. in terms nod
3. Solve the resulting simultaneous equations to obtain the unknown node voltage

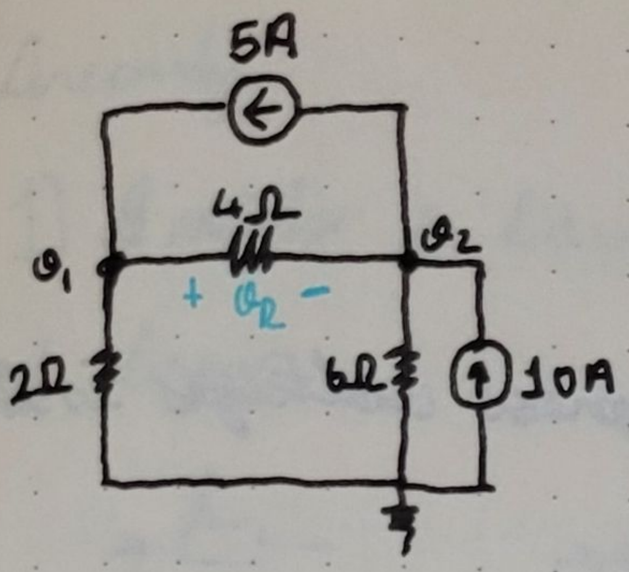


KCL @ node 1: $I_1 = I_2 + i_3 + i_4$

$$i_3 \cdot R_2 = v_1 - v_2 \Rightarrow i_3 = \frac{v_1 - v_2}{R_2}$$

$$i_4 \cdot R_1 = v_1 - 0 \Rightarrow i_4 = \frac{v_1}{R_1}$$

$$I_1 = I_2 + \frac{v_1 - v_2}{R_2} + \frac{v_1}{R_1} \qquad I_2 + \frac{v_1 - v_2}{R_2} = \frac{v_2 - 0}{R_3}$$

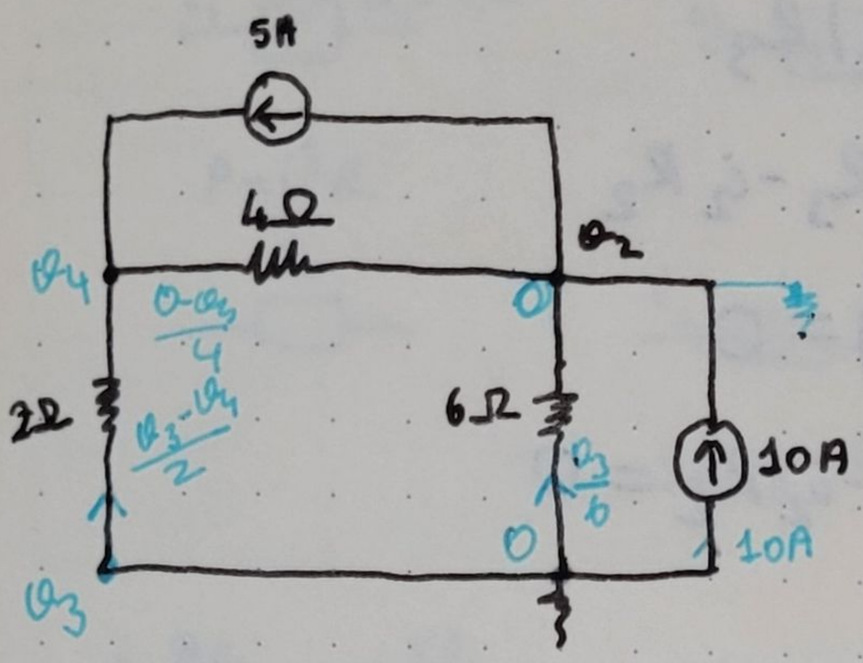


KCL1: $5 + \frac{v_2 - v_1}{4} = \frac{v_1}{2} \rightarrow 20 + v_2 - v_1 = 2v_1$

KCL2: $10 = 5 + \frac{v_2 - v_1}{4} + \frac{v_2}{6} \rightarrow 60 = 3v_2 - 3v_1 + 2v_2$

$\rightarrow v_1 = \frac{40}{3} V$, $v_2 = 20V$

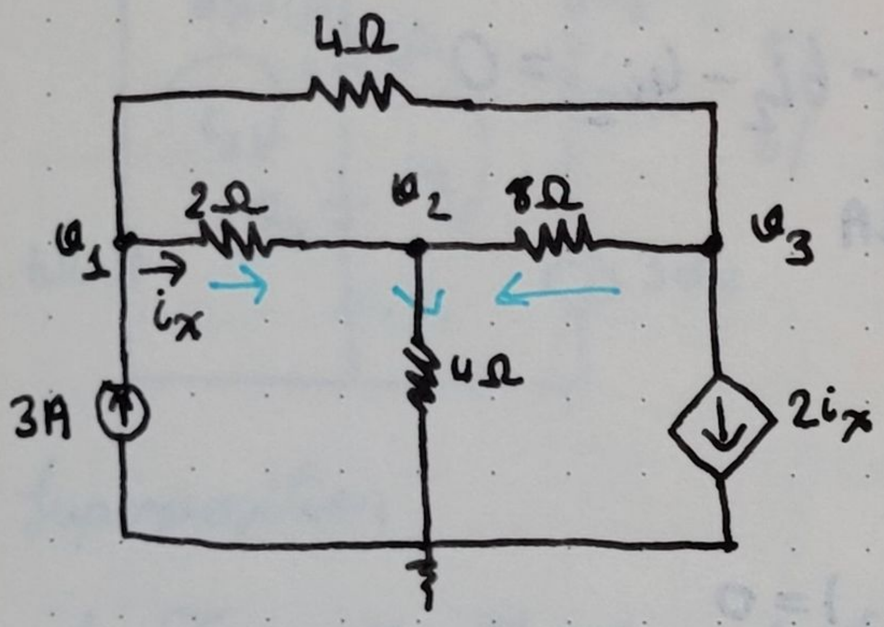
$v_2 = v_1 - v_2 = \frac{40}{3} - 20 = \frac{-20}{3}$



$5 - \frac{v_4}{4} + \frac{v_3 - v_4}{2} = 0$

$10 + \frac{v_3}{6} + \frac{v_3 - v_4}{2} = 0$

$\Rightarrow v_3 = -20V$, $v_4 = \frac{-20}{3} V$



$i_x = \frac{v_1 - v_2}{2}$

KCL1: $3 + \frac{v_3 - v_1}{4} = \frac{v_1 - v_2}{2}$

KCL2: $\frac{v_1 - v_2}{2} + \frac{v_3 - v_2}{8} = \frac{v_2}{4}$

$12 + v_3 - v_1 = 2v_1 - 2v_2$

$4v_1 - 4v_2 + v_3 - v_2 = 2v_2$

$\frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = v_1 - v_2$

$\frac{v_3 - v_1}{4} + \frac{v_3 - v_2}{8} = -2 \cdot \left(\frac{v_1 - v_2}{2} \right)$

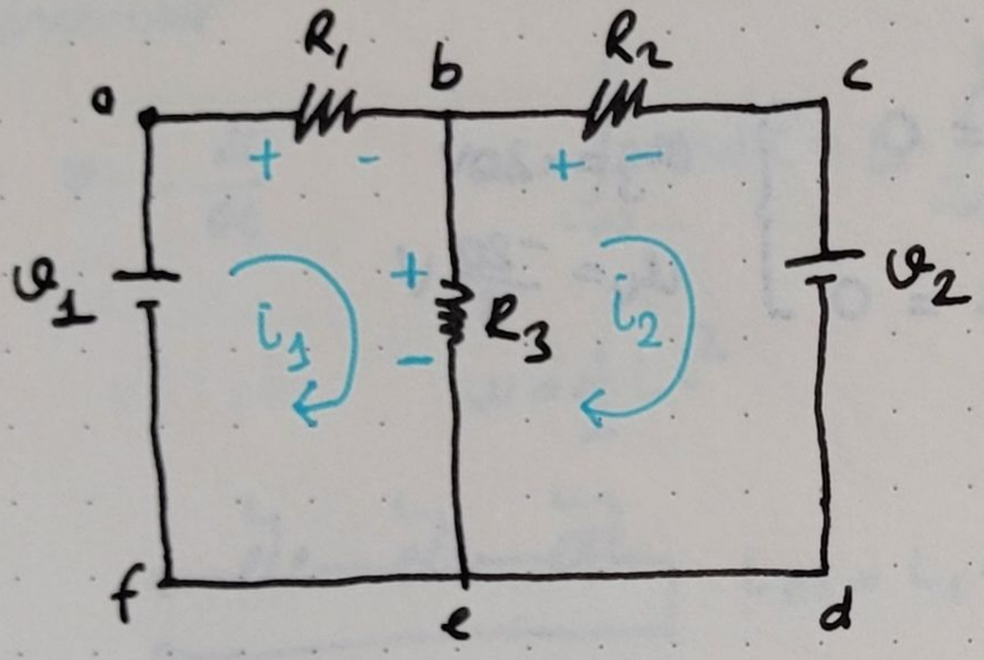
$2v_1 - 2v_3 + v_2 - v_3 = 8v_1 - 8v_2$

Mesh Analysis

1. Assign mesh currents i_1, i_2, i_n to the n meshes.

2. Apply KVL to all meshes. Use Ohm's Law to express voltages in terms of mesh currents.

3. Solve the resulting n simultaneous equations to get mesh currents.

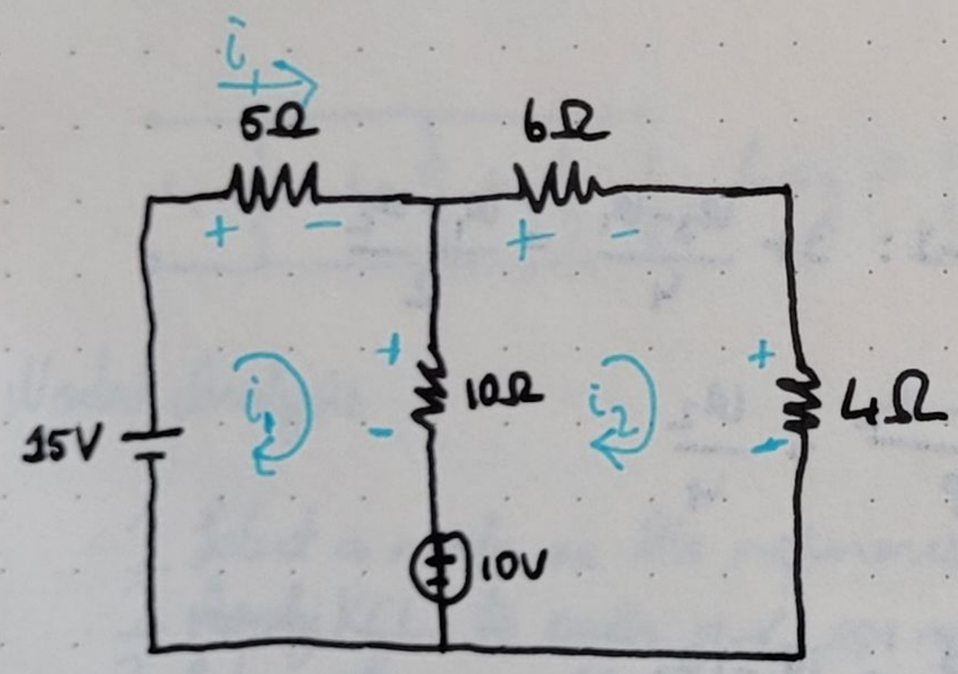


$$0 = v_1 - i_1 R_1 - (i_1 - i_2) R_3$$

$$0 = -v_2 + (i_1 - i_2) R_3 - i_2 R_2$$

$$v_1 - i_1 R_1 - (i_1 - i_2) = 0$$

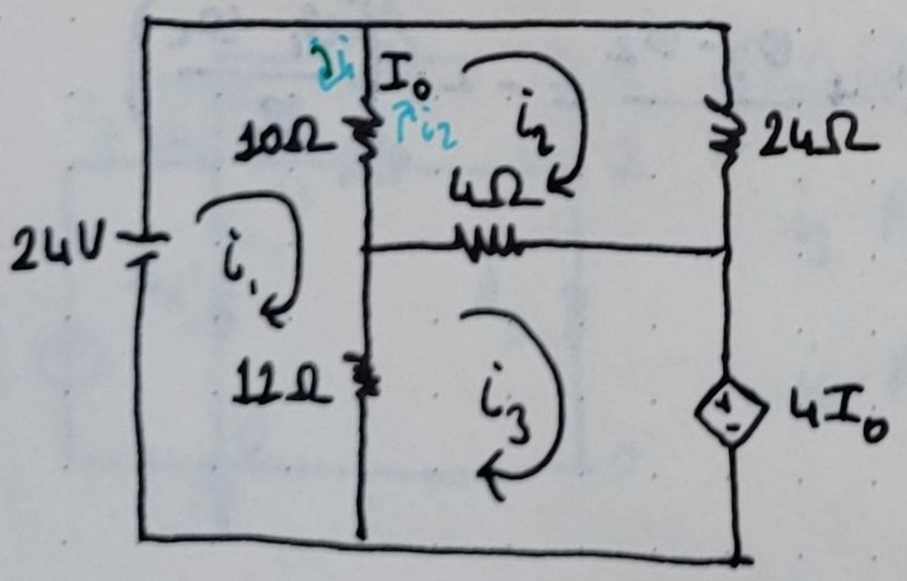
$$-v_2 - R_3 (i_2 - i_1) - i_2 R_2 = 0$$



$$15 - 5i_1 - (i_1 - i_2)10 - 10 = 0$$

$$10V - 10(i_2 - i_1) - 6i_2 - 4i_2 = 0$$

$$i_1 = 1A, \quad i_2 = 1A$$



$$24 - 10(i_1 - i_2) - 12(i_1 - i_3) = 0$$

$$-24 i_2 - 4(i_2 - i_3) - 10(i_2 - i_1) = 0$$

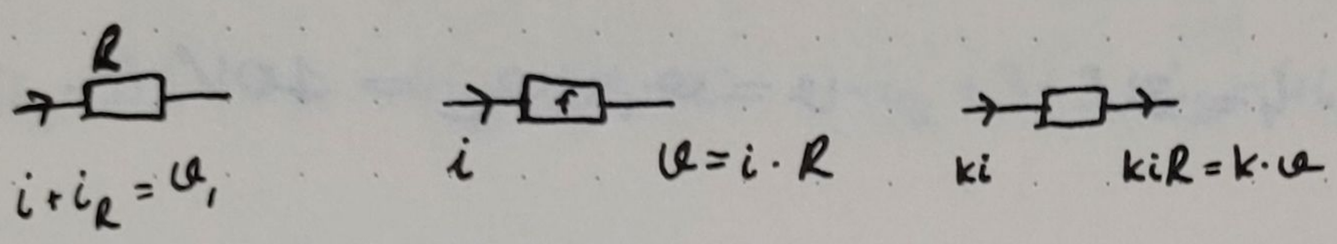
$$-4I_o - 12(i_3 - i_1) - 4(i_3 - i_2) = 0$$

$$-22 i_1 + 10 i_2 + 12 i_3 = -24$$
$$10 i_1 - 38 i_2 + 4 i_3 = 0$$
$$8 i_1 + 8 i_2 - 16 i_3 = 0$$

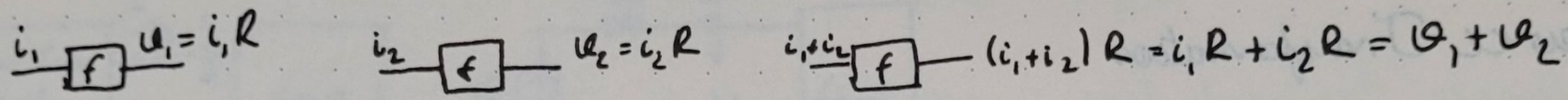
$$\begin{bmatrix} -22 & 10 & 12 \\ 10 & -38 & 4 \\ 8 & 8 & -16 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -24 \\ 0 \\ 0 \end{bmatrix}$$

Linearity

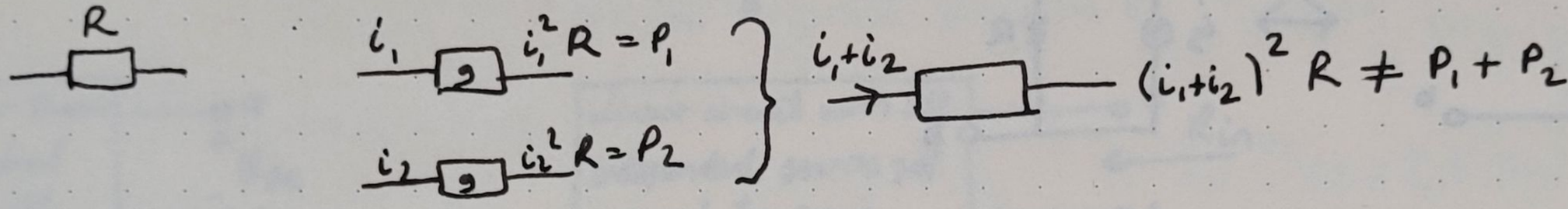
1) A resistor is linear. Relation between input current and output power is not linear.

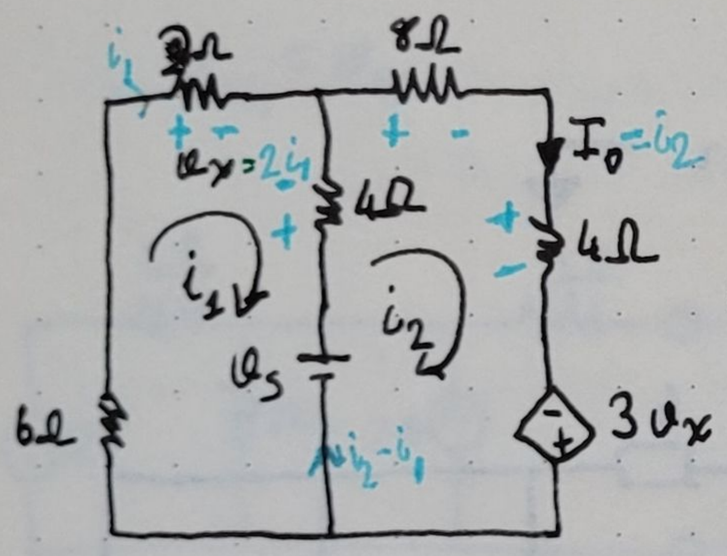


2)



$P = i^2 R$





$-u_s - 6i_1 - 2i_1 - 4(i_2+i_2) = 0 \quad u_s - 4(i_2 - i_1) - 8i_2 - 4i_2 + 6i_1 = 0$

$$\left.\begin{array}{l} 5/\ -12 i_1 + 4 i_2 = 24 u_s \\ 6/\ \ 10 i_1 - 16 i_2 = -24 - u_s \end{array}\right\} \quad -76 i_2 = -224 \rightarrow i_2 = 56/19$$

Superposition

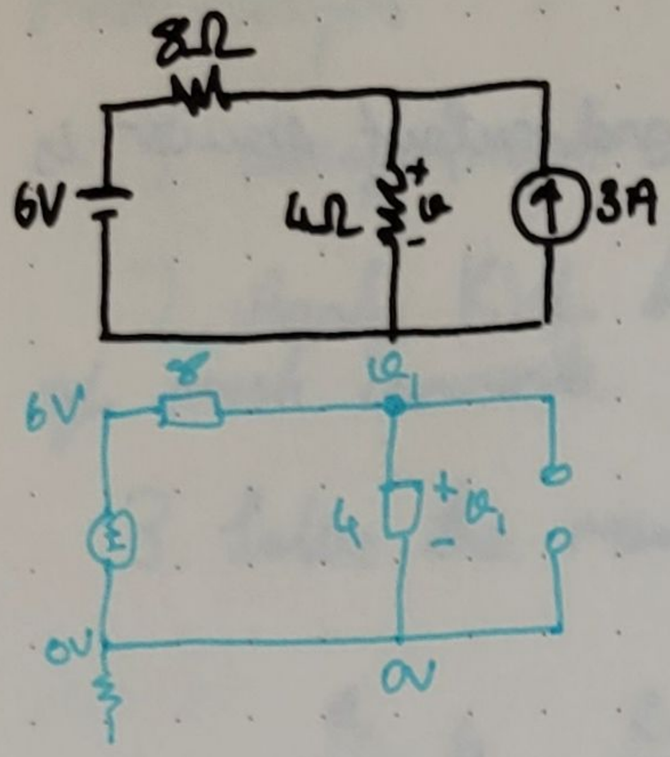
1. Turn off all independent resources.

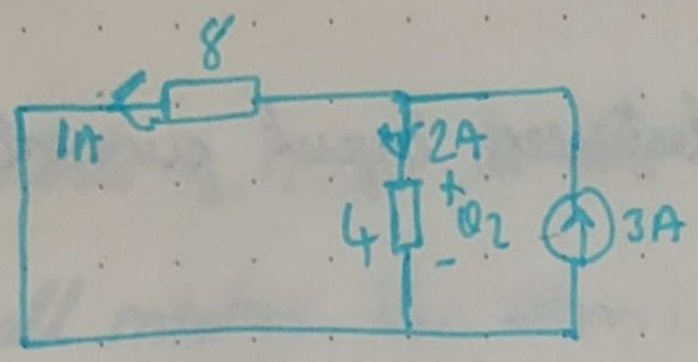
   - Replace voltage source with 0V (short circuit)
   - Replace current source with 0A (open circuit)

   Find the output (voltage or current) due to that active source.

2. Repeat step 1 for all independent sources.

3. Find the total contribution by adding algebraically all the contributions due to independent sources.





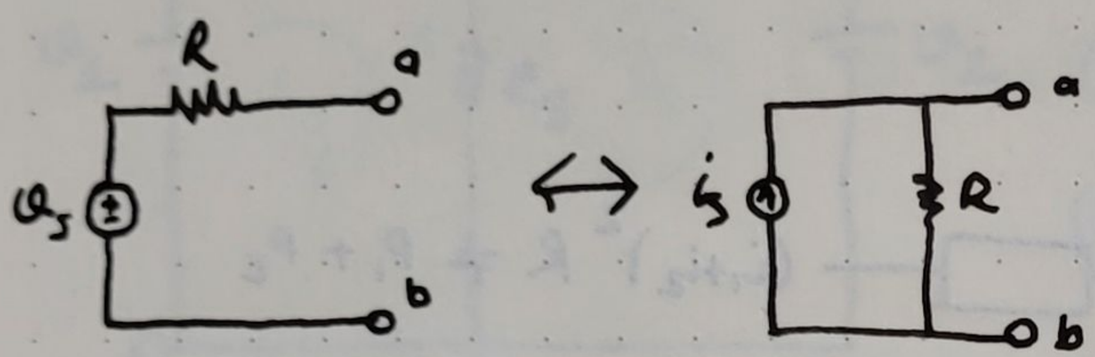
$$v_1 = \frac{4}{8+4} \cdot 4 = 2V$$

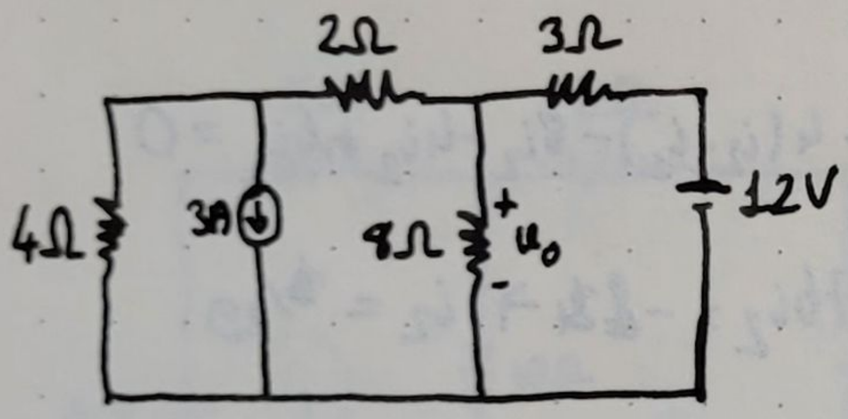
$$i_2 = \frac{3}{8+4} \cdot 8 = 2A$$

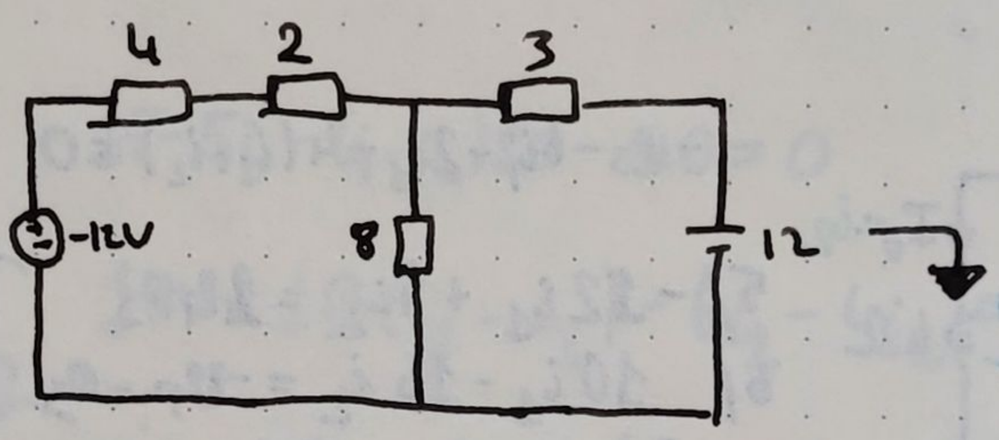
$$v_2 = i_2 \cdot 4 = 8V$$

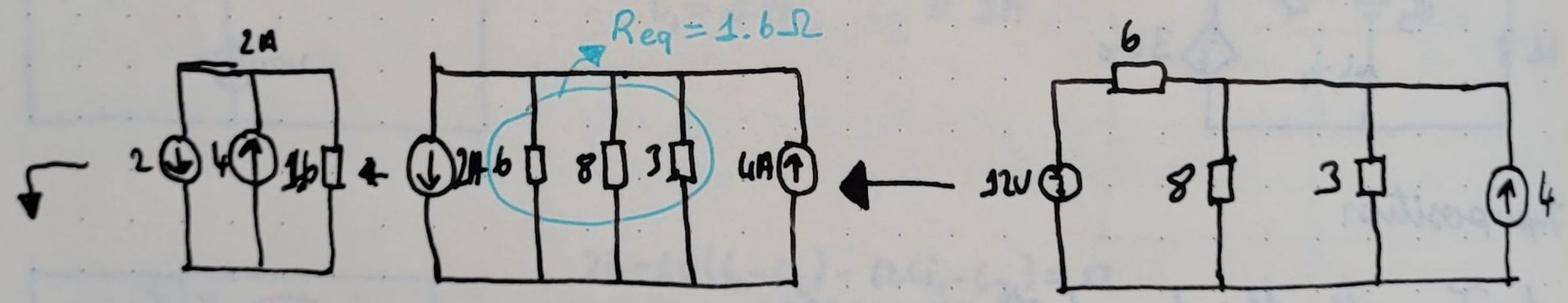
$$v = v_1 + v_2 = 10V$$

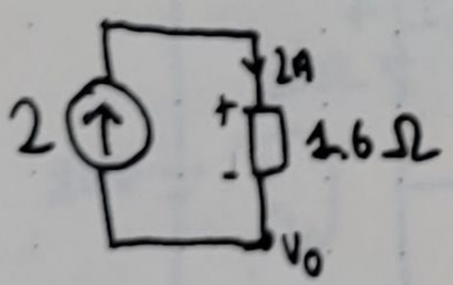
Source Transformation

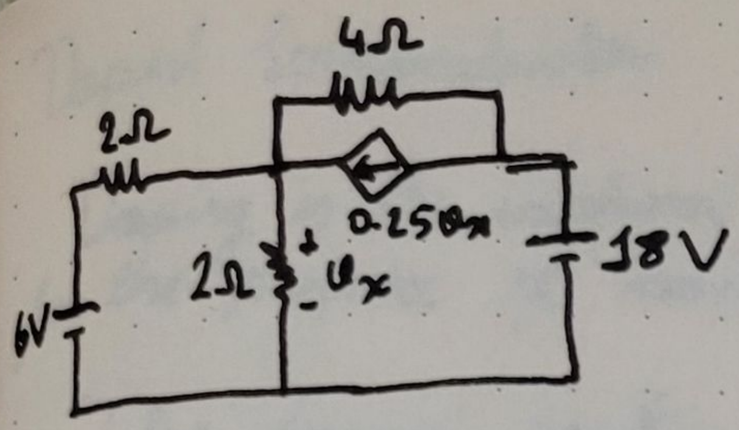


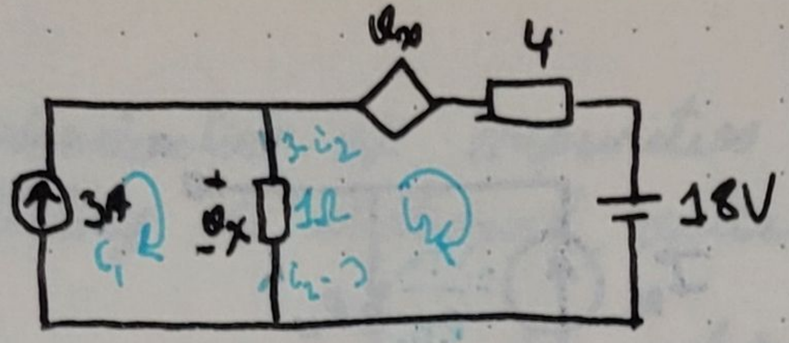










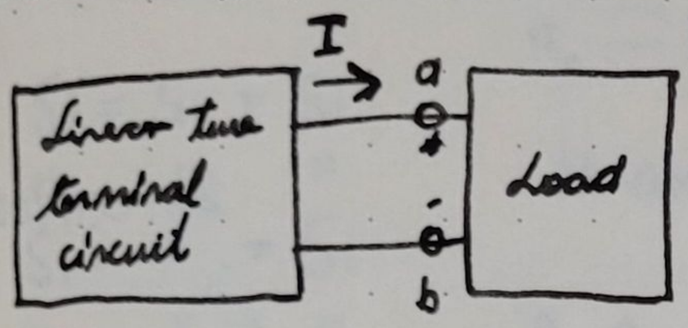


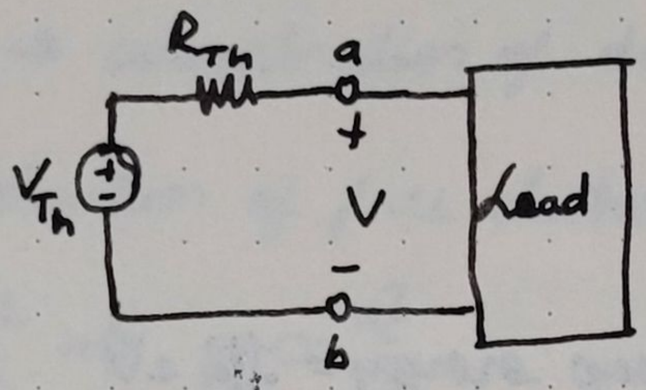
$v_x = (3-i_2) = 3-\left(\frac{-3}{2}\right) = 15/2$

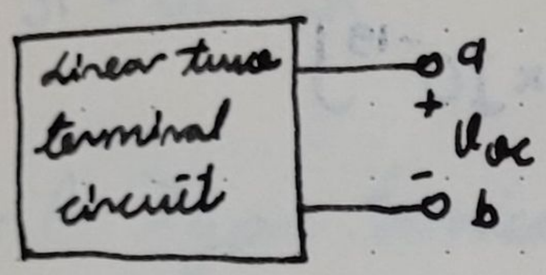
$i_1 = 3A$

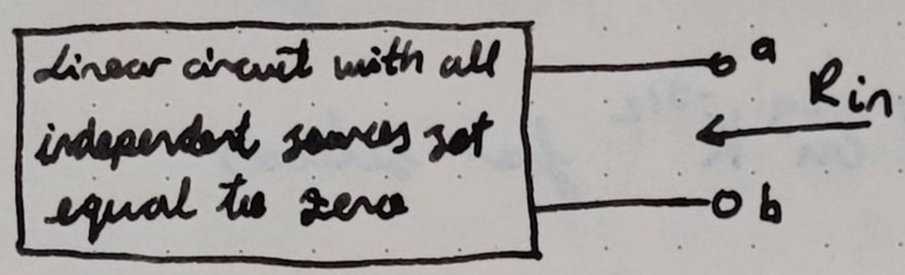
$-(i_2-3)1 - v_x - 4i_2 - 18 = 0 \qquad -i_2+3-3+i_2-4i_2-18=0 \rightarrow i_2 = \frac{-9}{2}$

Thevenin's Theorem



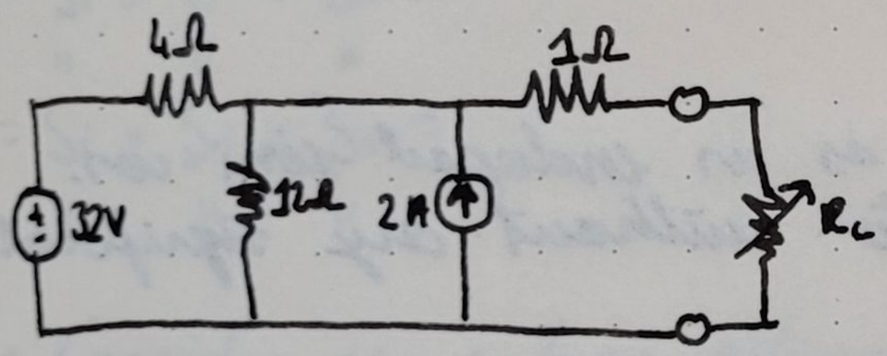


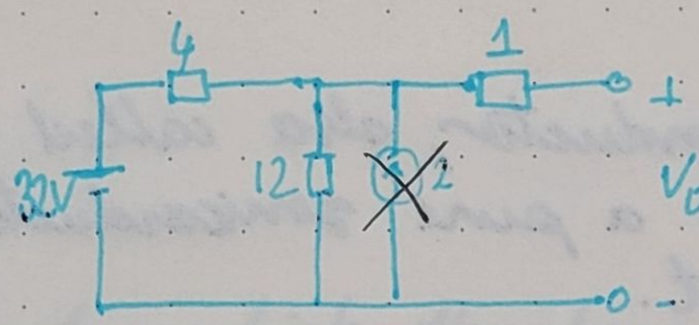




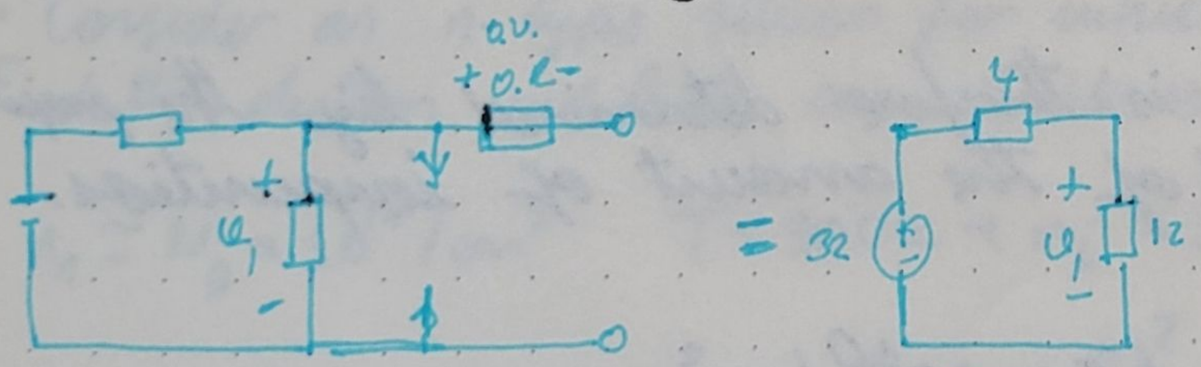
$V_{Th} = v_{oc}$

$R_{Th} = R_{in}$



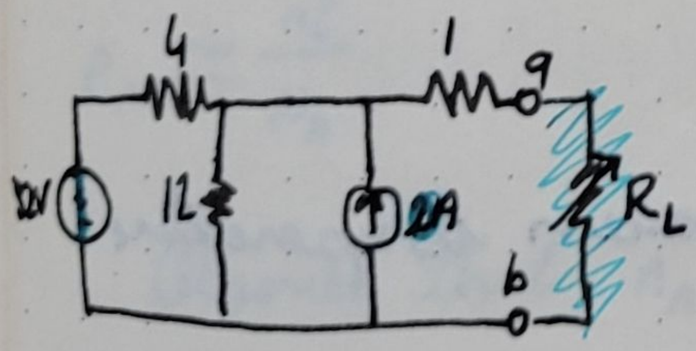


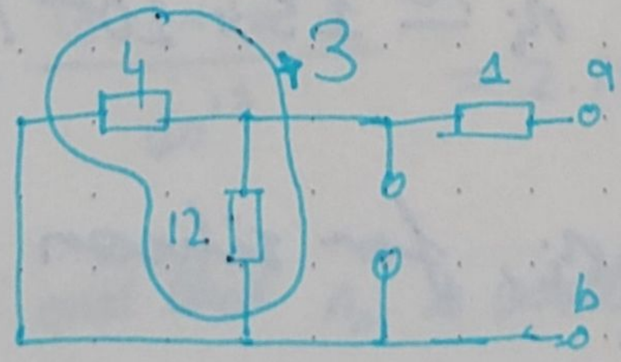
$V_{oc} = V_{Th}$



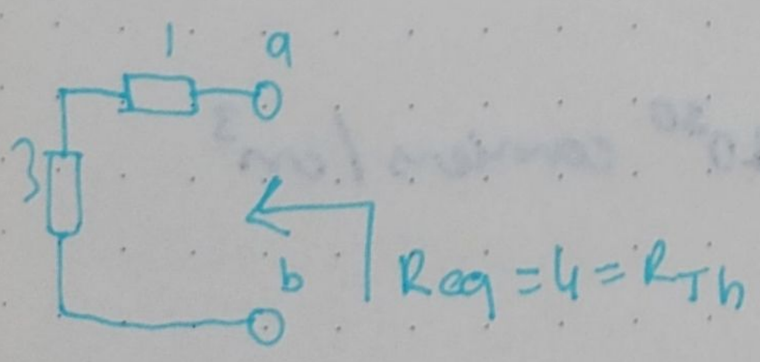
$v_1 = \frac{32}{16} \cdot 12 = 24V$

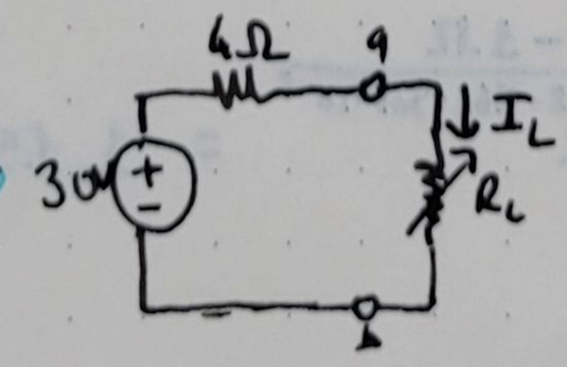
$V_{oc} = 24V = V_{Th}$



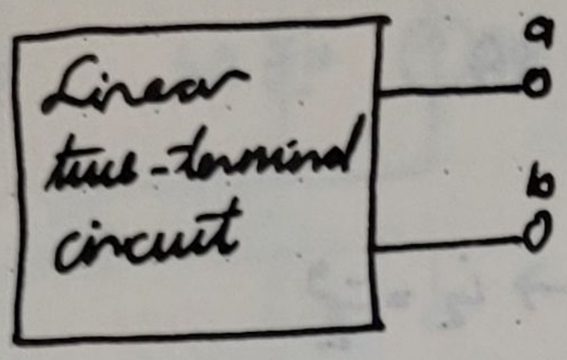


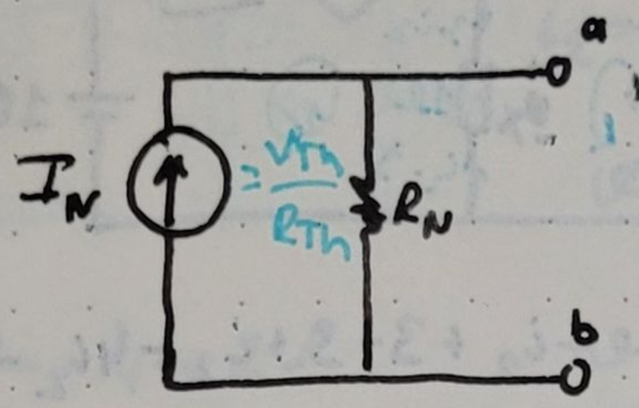
$R_{Th} = R_{eq}$





Norton's Theorem





$R_N = R_{Th}$ $\quad I_N = \frac{V_{Th}}{R_{Th}}$

Intrinsic Semiconductors

28/03/2021

$n_i = B \cdot T^{3/2} e^{-E_g/2kT}$ → Bandgap energy = 1.12 eV

→ $7.3 \times 10^{15}\ cm^{-3} K^{-3/2}$ for silicon

$1eV = 1.6 \times 10^{-19} J$

$n = p = n_i$ → denotes number of free electrons and holes in a unit volume (cm^3) of intrinsic silicon

→ n free electrons / cm^{-3}

→ p holes / cm^{-3}

An intrinsic (pure) semiconductor, also called as an undoped semi-con. or i-type semi con., is a pure semiconductor without any significant dopant species present.

The number of charge carries is therefore determined by the properties of the materials itself instead of the amount of impurities.

$pn = n_i^2$ for silicon $n_i \simeq 1.5 \times 10^{10} / cm^3$

- Calculate the value of n_i, for silicon at room temperature ($T \simeq 300K$)

$n_i = (7.3 \times 10^{15}) \cdot 300^{\frac{3}{2}} \cdot e^{\frac{-1.12}{2 \cdot 8.62 \cdot 300 \cdot 10^{-5}}} = 1.15 \times 10^{10}$ carriers / cm^3

Doped Semiconductors

Doping is the intentional introduction of impurities into an intrinsic semicon. for the purpose of modulating its electrical, optical and structural prope.

The doped material is referred to extrinsic semicon.

- N-Type Semiconductor

Si = Si = Si
" " "
Si = P = Si
" v "
Si = Si = Si

$n_n \simeq N_D$ → concentration of donor atoms

↳ concentration of free electrons

$p_n n_n = n_i^2 \rightarrow p_n \simeq \frac{n_i^2}{N_D}$

- P-Type Semiconductor

= Si = Si = Si =
" " D
= Si = B = Si =
" " "
= Si = Si = Si =
" " "

$p_p \simeq N_A \quad p_p n_p = n_i^2 \rightarrow n_p \simeq \frac{n_i^2}{N_A}$

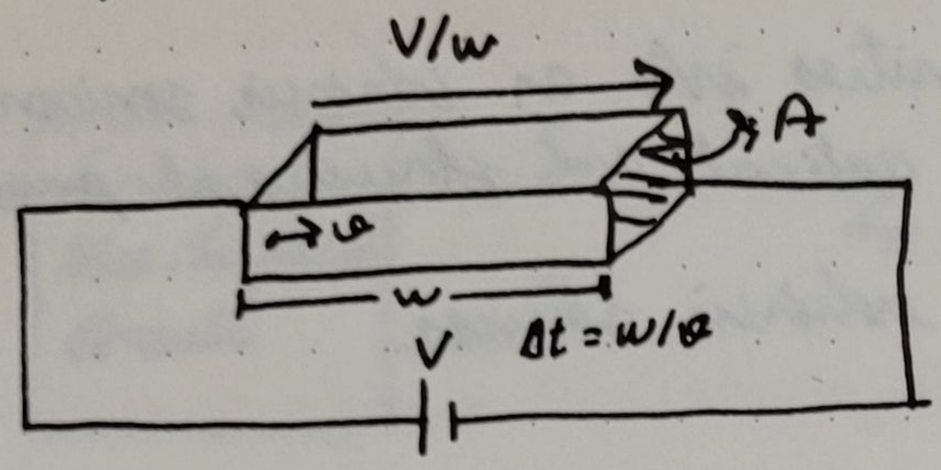
• Consider an n-type silicon for which the dopant concentration $N_D = 10^{17}/cm^3$
Find the electron and hole concentrations at $T = 300K$

$n_n \simeq N_D = 10^{17}/cm^3 \quad T = 300K \rightarrow n_i = 1.5 \times 10^{10}/cm^3$

$p_n \simeq \frac{n_i^2}{N_D} \qquad p_n = \frac{(1.5 \times 10^{10})^2}{10^{17}} = 2.25 \times 10^3/cm^3$

Observe that $n_n \gg n_i$ and that n_n is vastly higher than p_n.

Drift Current



$$v_p = E \cdot \mu_p$$

$$\Delta Q = w \cdot A \cdot \rho \cdot q \quad (\rho \rightarrow \text{\# of holes})$$

$$I = \frac{\Delta Q}{\Delta t} = \frac{wA\rho q}{w/v} = A \cdot \rho \cdot q \cdot E \cdot \mu_p$$

$$I/A = J = \rho q \cdot E\mu_p = E\underbrace{(q_p \mu_p - q_n \mu_n)}_{\sigma} \qquad \sigma^{-1} = \rho$$

$$R = \frac{\rho \cdot l}{A}$$

$$J = \sigma E \Rightarrow \frac{I}{A} = \sigma \cdot \frac{V}{w} \Longrightarrow V = \frac{1}{\sigma} \cdot \frac{w}{A} \cdot I$$

- Find the resistivity of (a) intrinsic silicon and (b) p-type silicon with $N_A = 10^{16}/cm^3$. Use $n_i = 1.5 \times 10^{10}/cm^3$, and assume that for intrinsic silicon $\mu_n = 1350 cm^2/V \cdot s$ and $\mu_p = 480 cm^2/V \cdot s$, and for the doped silicon $\mu_n = 1110 cm^2/V \cdot s$ and $\mu_p = 400 cm^2/V \cdot s$. (Note that doping results in reduced carrier mobilities)

(a) For intrinsic silicon,

$$p = n = n_i = 1.5 \times 10^{10}/cm^3$$

Thus,

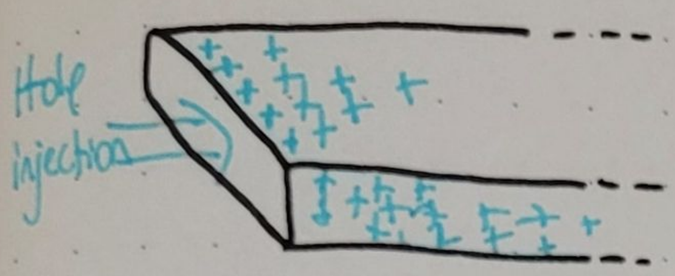
$$\rho = \frac{1}{q(p\mu_p + n\mu_n)} = 2.28 \times 10^5 \, \Omega \cdot cm$$

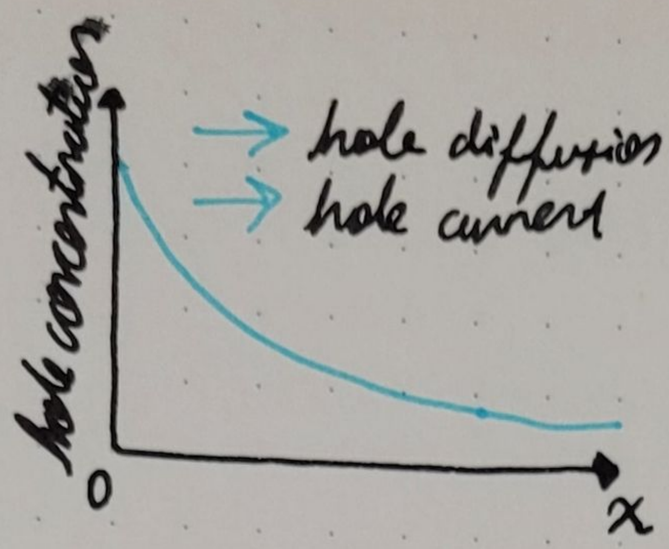
(b) For p-type silicon,

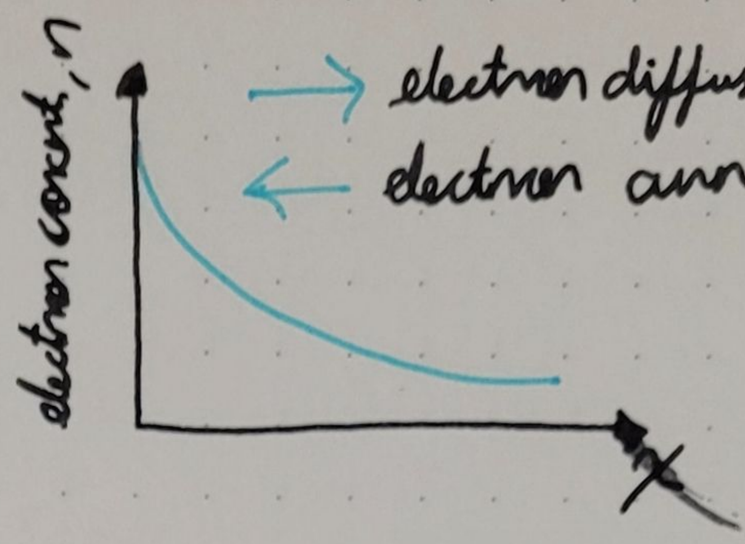
$$p_p \simeq N_A = 10^{16}/cm^3 \qquad n_p \simeq \frac{n_i^2}{N_A} = \frac{(1.5 \times 10^{10})^2}{10^{16}} = 2.25 \times 10^4/cm^3$$

$$\rho = \frac{1}{q(p\mu_p + n\mu_n)} = 1.56 \, \Omega cm$$

Diffusion Current







- $J_p = -qD_p \frac{dp(x)}{dx}$

$J_n = qD_n \frac{dn(x)}{dx}$

- Consider a bar of silicon in which a hole concentration profile described by $p(x) = p_0 e^{-x/L_p}$ is established. Find the hole-current density at $x=0$. Let $p_0 = 10^{16}/cm^3$, $L_p = 1\mu m$ and $D_p = 12 cm^2/s$. If the cross-sectional area of the bar is $100\mu m^2$, find the current I_p.

$$J_p = -qD_p \frac{dp(x)}{dx} = -qD_p \frac{d}{dx}\left[p_0 e^{-x/L_p}\right] = q \cdot \frac{D_p}{L_p} p_0 e^{-x/L_p}$$

$$J_p(0) = q\frac{D_p}{L_p} p_0 = 1.6\times10^{-19} \times \frac{12}{1\times10^{-4}} \times 10^{16} = 192\ A/cm^2$$

$$I_p = J_p \times A = 192 \times 100 \times 10^{-8} = 192\mu A$$